

☒ Easy (10 Questions)

Q1. Which firewall operates by inspecting only packet headers?

- A) Application Gateway
- B) Packet Filter
- C) Stateful Firewall
- D) NGFW

Answer: B

☒ *Packet filters analyze IP, port, and protocol without tracking session.*

Q2. What is the primary function of iptables?

- A) DNS resolution
- B) File permission management
- C) Packet filtering in Linux
- D) Memory allocation

Answer: C

❑ *iptables is used to configure Linux-based firewall rules.*

Q3. Which firewall tracks active connections and allows only related traffic?

- A) Proxy
- B) Packet Filter
- C) Stateful Inspection
- D) Static Filter

Answer: C

❑ *Stateful firewalls keep a state table and monitor traffic flow context.*

Q4. In a screened host firewall architecture, what is the role of the bastion host?

- A) Backup storage
- B) Application-layer inspection
- C) Print server
- D) DNS caching

Answer: B

Q5. Which command sets the default policy for incoming traffic to DROP in iptables?

- A) iptables --input drop
- B) iptables -P INPUT DROP
- C) iptables -I drop INPUT
- D) iptables INPUT reject

Answer: B

Q6. Which firewall provides application awareness?

- A) Packet filter
- B) Proxy firewall
- C) Stateful firewall
- D) Next-Gen Firewall

Answer: D

Q7. What layer does a proxy firewall operate at?

- A) Layer 2
- B) Layer 3
- C) Layer 7
- D) Layer 5

Answer: C

Q8. Which iptables chain is used for traffic being forwarded through the machine?

- A) INPUT
- B) OUTPUT
- C) FORWARD
- D) NAT

Answer: C

Q9. Which of the following is true about NGFW?

- A) Can only inspect Layer 3 traffic
- B) Tracks only TCP packets
- C) Can identify and control applications
- D) Replaces switches in LAN

Answer: C

Q10. Which Linux firewall utility was designed to replace iptables?

- A) netcut
- B) nftables
- C) pfSense
- D) iproute2

Answer: B

☒ **Medium and Hard MCQs (30 remaining) will follow in the next message...**

Shall I continue?

You said:

yes

ChatGPT said:

Continuing with the remaining:

**☒ Packet Filtering, Stateful Inspection, Screened Host, NGFW, iptables
– MCQs (continued)**

☒ Medium (20 Questions)

Q11. In iptables, which **table** is responsible for NAT operations like masquerading?

- A) filter
- B) nat
- C) mangle
- D) raw

Answer: B

☒ *The nat table is used for destination NAT (DNAT), source NAT (SNAT), and masquerading.*

Q12. What makes stateful inspection more secure than packet filtering?

- A) Tracks operating system behavior
- B) Inspects TCP flags and connection states
- C) Works at Layer 2
- D) Requires more RAM

Answer: B

☒ *Stateful firewalls maintain the state of active connections.*

Q13. The key component of a **screened host firewall** is:

- A) Encrypted database
- B) Transparent proxy
- C) Bastion host
- D) DMZ router

Answer: C

Q14. What does NGFW use to detect threats in encrypted traffic?

- A) DMZ filtering
- B) Static routing
- C) TLS/SSL inspection
- D) Port blocking

Answer: C

☒ *TLS/SSL inspection allows NGFWs to decrypt and inspect encrypted traffic.*

Q15. Which iptables rule would **allow** all **outgoing HTTP** traffic?

- A) iptables -A OUTPUT -p tcp --dport 80 -j ACCEPT
- B) iptables -A INPUT -p tcp --dport 80 -j DROP
- C) iptables -A FORWARD -p tcp --sport 80 -j ACCEPT
- D) iptables -D OUTPUT -p tcp --dport 80 -j ACCEPT

Answer: A

Q16. Which chain in iptables handles traffic sent **from** the local system?

- A) INPUT
- B) OUTPUT
- C) FORWARD
- D) NAT

Answer: B

Q17. Which of the following is **not** a typical NGFW feature?

- A) Deep packet inspection
- B) Application filtering
- C) MAC address spoofing
- D) Intrusion prevention

Answer: C

☒ *MAC spoofing is an attack, not a NGFW feature.*

Q18. A **bastion host** should be:

- A) A user desktop
- B) Unpatched and open
- C) Hardened and monitored
- D) Behind NAT and firewall

Answer: C

Q19. Which of the following best explains why packet filters are vulnerable to spoofing?

- A) They are too slow
- B) They only inspect IP and port, not state
- C) They encrypt traffic by default
- D) They inspect application payloads

Answer: B

Q20. What makes NGFWs different from traditional firewalls?

- A) They only use IP filtering
- B) They track user behavior and app-level traffic
- C) They ignore TCP ports
- D) They use dynamic DNS rules

Answer: B

Q21. Which command removes all iptables rules from the INPUT chain?

- A) iptables -R INPUT
- B) iptables --flush INPUT
- C) iptables -F INPUT
- D) iptables -Z INPUT

Answer: C

☒ *-F or --flush clears rules from a chain.*

Q22. Which protocol does a stateful firewall analyze to track the session state?

- A) UDP
- B) ICMP
- C) TCP
- D) SMTP

Answer: C

☒ *TCP has session states like SYN, ACK, FIN, etc.*

Q23. Why are NGFWs ideal in Zero Trust architecture?

- A) They allow all traffic
- B) They identify and control users and applications
- C) They eliminate logging
- D) They rely on MAC filtering

Answer: B

Q24. Which iptables command lists active rules?

- A) iptables -L
- B) iptables -A
- C) iptables -N
- D) iptables -Z

Answer: A

Q25. Which of the following can detect and block threats in encrypted HTTPS traffic?

- A) Stateless firewall
- B) DNS filter
- C) NGFW with SSL inspection
- D) Basic NAT router

Answer: C

Q26. Which command **drops all** incoming **ICMP echo requests**?

- A) iptables -A INPUT -p tcp --dport 7 -j DROP
- B) iptables -A INPUT -p icmp --icmp-type echo-request -j DROP
- C) iptables -D FORWARD icmp
- D) iptables -P ICMP DROP

Answer: B

Q27. In iptables, a **chain** is a:

- A) Policy file
- B) Group of network devices
- C) Rule set that matches packets in a certain context
- D) VPN tunnel

Answer: C

Q28. A firewall configured to allow outbound traffic only if a connection was initiated from inside is using:

- A) Packet filtering
- B) Proxy forwarding
- C) Stateful inspection
- D) Reverse DNS

Answer: C

Q29. Application-layer filtering requires the firewall to:

- A) Operate at Layer 2
- B) Analyze only IP addresses
- C) Understand application protocols
- D) Be connected to a NAT router

Answer: C

Q30. What does the following command do?

```
bash
```

Copy code

```
iptables -A INPUT -m conntrack --ctstate ESTABLISHED,RELATED -j ACCEPT
```

- A) Blocks all related connections
- B) Accepts new connections only
- C) Accepts packets that are part of or related to an established connection
- D) Drops all packets

Answer: C

☒ Hard (10 Questions)

Q31. What is a **limitation** of stateful firewalls?

- A) Cannot filter port numbers
- B) Cannot inspect encrypted payloads
- C) Cannot track TCP handshakes
- D) Cannot detect IP addresses

Answer: B

☒ *Encrypted content needs SSL inspection (feature of NGFW).*

Q32. What is the role of `--dport` in an iptables rule?

- A) Sets default input policy
- B) Matches destination port
- C) Specifies DNS entry
- D) Logs rejected packets

Answer: B

Q33. What would this iptables command do?

bash

Copy code

```
iptables -A FORWARD -p tcp --dport 443 -j ACCEPT
```

- A) Block HTTPS on the local system
- B) Allow HTTPS traffic through the router
- C) Reject port 443 for INPUT
- D) Log HTTPS traffic

Answer: B

Q34. Which of the following best defines **Deep Packet Inspection (DPI)**?

- A) Filtering by MAC address
- B) Analyzing header fields only
- C) Examining packet payload and header
- D) Scanning routing tables

Answer: C

Q35. How can NGFWs enforce identity-based security?

- A) By using only MAC addresses
- B) By analyzing usernames from Active Directory
- C) By counting HTTP packets
- D) By setting BIOS-level passwords

Answer: B

Q36. What is a **stateless** firewall unable to do?

- A) Block UDP traffic
- B) Log activity
- C) Track connection state
- D) Match IP addresses

Answer: C

Q37. Which of the following iptables chains is used when a Linux box is **routing traffic** (acting as a gateway)?

- A) INPUT
- B) OUTPUT
- C) FORWARD
- D) PREROUTING

Answer: C

Q38. NGFWs often include IPS functionality. What does IPS stand for?

- A) Internal Packet System
- B) Intrusion Prevention System
- C) Internal Proxy Server
- D) Internet Protocol Stack

Answer: B

Q39. What is the advantage of combining proxy firewall and NGFW?

- A) Ignores TCP flags
- B) Allows only port 80
- C) Enables both DPI and session caching
- D) Blocks MAC spoofing

Answer: C

Q40. Which iptables chain and table combination would you use to implement **SNAT**?

- A) OUTPUT in filter
- B) INPUT in mangle
- C) POSTROUTING in nat
- D) FORWARD in raw

Answer: C

☒ *SNAT uses POSTROUTING in the nat table to modify the source IP.*

